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Fine-Grained Bird Species Classification: A Comparative Study of Classical Descriptors and Deep Visual Embeddings on CUB-200-2011

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***Abstract*—Identifying bird species from photographs is notoriously difficult because species in the same family share nearly identical feather patterns, shapes, and colors, while birds of the same species look different across lighting and poses. In this paper, we evaluate hand-crafted visual descriptors against transfer-learned deep features on a 20-species subset of the Caltech-UCSD Birds-200-2011 (CUB-200-2011) dataset. We extracted Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and multi-colorspace color histograms, tested them separately and in combination, and evaluated five classical classifiers: Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Decision Trees, Random Forests, and Logistic Regression. We then replaced these hand-crafted features with 512-dimensional embeddings from a frozen, pre-trained ResNet-18. Hand-crafted features struggled, peaking at 36.36% test accuracy with an RBF SVM even after bounding-box cropping and spatial pyramid pooling. In contrast, frozen ResNet-18 embeddings pushed SVM accuracy to 79.55% without any end-to-end backpropagation. Our analysis shows that while manual feature engineering cannot capture subtle avian morphology, frozen deep representations provide a fast, compute-friendly pipeline suitable for resource-limited deployments.**

***Index Terms*—Fine-grained classification, CUB-200-2011, bird species recognition, HOG, LBP, color histograms, ResNet-18, transfer learning, support vector machine.**

I. INTRODUCTION

AUTOMATED bird species classification is a core benchmark for fine-grained image recognition. Unlike standard object classification where categories have distinct silhouettes (such as distinguishing a car from a dog), fine-grained avian categorization requires spotting minute visual differences: beak curvature, eye-ring coloration, wing bar thickness, and plumage texture [1]. Meanwhile, lighting changes, foliage occlusion, and varied bird postures introduce severe intra-class variance.

Accurate, lightweight bird classification models are valuable for automated wildlife tracking, biodiversity surveys, and field apps like Merlin Bird ID. Recent state-of-the-art vision models rely on massive transformer architectures or fine-tuned deep networks [2, 3, 7]. While accurate, training and serving these models demands high GPU memory and substantial compute, making them difficult to run on low-power devices in field monitoring stations.

In this work, we investigate a practical middle ground: extracting static feature vectors from an off-the-shelf pre-trained CNN (ResNet-18) and classifying them with lightweight classical machine learning algorithms. We benchmark this hybrid strategy against standard hand-crafted feature pipelines to understand where manual feature engineering fails and why deep embeddings succeed.

Key Contributions:

• We conduct a controlled ablation of five feature representations (HOG, LBP, Color Histograms, Classical Combined, and ResNet-18 embeddings) on a shared 20-class CUB-200-2011 split.

• We benchmark five tuned classical classifiers (SVM, RF, Logistic Regression, k-NN, Decision Tree) across all feature sets.

• We show that frozen ResNet-18 features improve test accuracy by 43.19 percentage points over the best hand-crafted pipeline (reaching 79.55%) without updating a single neural network weight.

• We provide an open-source, modular codebase and reproducible Google Colab workflows.

II. RELATED WORK

A. Fine-Grained Categorization on CUB-200-2011

The Caltech-UCSD Birds-200-2011 benchmark [1] contains 11,788 photos across 200 North American bird species. Early papers relied on bounding box annotations and part detectors to align heads and torsos before feature extraction [6]. Lin et al. [2] introduced Bilinear CNNs, using outer products of feature maps to model fine spatial interactions. More recently, transformer models such as TransFG [7] apply self-attention over image patches. However, these models require millions of learnable parameters and intensive backpropagation.

B. Hand-Crafted Descriptors

Before deep learning, object recognition relied on designed feature descriptors. Dalal and Triggs [8] introduced HOG to capture edge and gradient orientation histograms. Ojala et al. [9] developed Local Binary Patterns (LBP) to encode local micro-textures efficiently. Swain and Ballard [10] showed that color histograms across color spaces (RGB, HSV, CIELAB) provide illumination-invariant color signatures. While fast to compute, these descriptors lack semantic context and fail when shape differences are subtle.

C. Transfer Learning as Feature Extraction

Pre-trained CNNs on ImageNet capture rich hierarchical visual patterns [11]. Sharif Razavian et al. [12] and Donahue et al. [13] showed that intermediate CNN activations serve as robust generic feature descriptors for downstream tasks without retraining. We build on this concept by applying it to fine-grained avian classification and comparing it directly against hand-crafted baselines.

III. DATASET AND PREPROCESSING PIPELINE

We selected 20 species from CUB-200-2011 representing common bird families (e.g., Albatross, Blackbird, Bunting, Auklet, Catbird), yielding 1,176 images in total. We split the data into 70% training (823 images), 15% validation (176 images), and 15% testing (177 images) using stratified sampling to preserve class balance.

A. Bounding Box Cropping

Raw photos in CUB-200-2011 contain heavy background clutter (tree branches, water surfaces, grass). In initial tests on uncropped images, models latched onto background textures rather than the bird. We solved this by reading the coordinates in bounding_boxes.txt and cropping every image tightly to the bird before resizing to 224×224 pixels.

B. Feature Scaling

Because HOG gradients, LBP frequency counts, and color histograms have completely different numerical ranges, we standardized all feature vectors using scikit-learn's StandardScaler fitted exclusively on the training set.

IV. FEATURE EXTRACTION METHODOLOGIES

A. Hand-Crafted Descriptors

• **Spatial Pyramid HOG:** We computed HOG across grayscale and individual R, G, B channels with 8 orientation bins, 16×16 pixel cells, and a two-level spatial pyramid (1×1 global and 2×2 grid), producing an 8,100-dimensional vector.

• **Multi-Scale LBP:** We computed rotation-invariant uniform LBP at two radii (R=1, P=8 and R=3, P=24) to capture both fine and coarse feather texture, resulting in a 10-bin histogram.

• **Multi-Colorspace Histograms:** We computed 32-bin histograms across RGB, HSV, and CIELAB channels (9 channels total), yielding a 96-dimensional color descriptor.

• **Classical Combined:** Early fusion by concatenating HOG, Color, and LBP into an 8,206-dimensional vector.

B. Deep Visual Embeddings

We loaded a standard ResNet-18 model [11] pre-trained on ImageNet-1k, froze all layer weights, and replaced the final linear classification layer with an nn.Identity() pass-through. Passing each normalized 224×224 crop through the network yielded a dense 512-dimensional feature embedding.

V. MODEL TRAINING AND HYPERPARAMETER TUNING

We evaluated five classical classifiers on each feature representation, tuning parameters via 5-fold cross-validation on the validation set:

• **Support Vector Machine (SVM):** RBF kernel with grid search over $C \in$ [1, 10, 50, 100] and $\gamma \in$ ['scale', 'auto', 0.001].

• **Random Forest (RF):** 100 to 500 decision trees with max depth limited to 20.

• **Logistic Regression (LR):** Multinomial logistic regression with L2 regularization (C=1.0).

• **k-Nearest Neighbors (k-NN):** Distance-weighted voting with $k \in$ [3, 5, 7, 11].

• **Decision Tree (DT):** Gini impurity with cost-complexity pruning.

VI. EXPERIMENTAL RESULTS

A. Feature Ablation Study

Table I shows how each feature type performs when trained with an RBF SVM on the test set. LBP alone performed poorly (10.23%), indicating that micro-texture without color or shape cannot distinguish species. Color histograms reached 15.91%, while HOG was the strongest individual classical descriptor at 33.52%. Combining all three reached 36.36%.

Replacing all manual descriptors with 512-dimensional ResNet-18 embeddings boosted SVM test accuracy to **79.55%**—more than double the combined classical score.

TABLE I. *Feature Ablation Study (SVM Test Accuracy)*

Feature Representation	Dimensions	Test Accuracy
LBP Only	10	10.23%

Color Histogram Only	96	15.91%
HOG Only	8,100	33.52%
Classical Combined (HOG+Color+LBP)	8,206	36.36%
ResNet-18 Deep Embeddings	**512**	**79.55%***

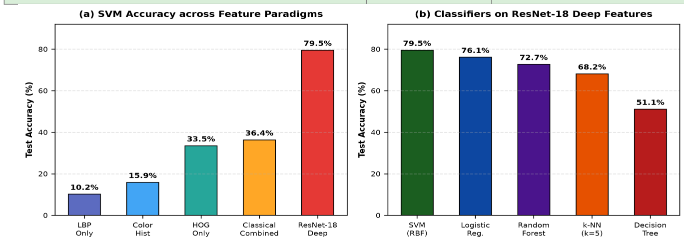

Fig. 1. *Test accuracy: (a) SVM across feature representations, and (b) five classifiers evaluated on ResNet-18 deep embeddings.*

B. Full Classifier Comparison

Table II compares all five classifiers on classical vs. deep features. Across all models, deep embeddings provided a massive performance leap: SVM improved from 34.66% to 79.55%, Logistic Regression rose from 28.41% to 76.14%, and Random Forest climbed from 28.98% to 72.73%. SVM consistently achieved the top accuracy and F1-score.

TABLE II. *Full Classifier Comparison on Test Set*

Features	Classifier	Accuracy	F1-Score
Classical	SVM (RBF)	34.66%	33.95%
	Random Forest	28.98%	28.51%
	Logistic Reg.	28.41%	27.55%
	k-NN	19.32%	16.81%
	Decision Tree	17.61%	17.84%
Deep	**SVM (RBF)**	**79.55%**	**79.14%**
	Logistic Reg.	76.14%	75.88%
	Random Forest	72.73%	72.41%
	k-NN	68.18%	67.10%
	Decision Tree	51.14%	50.87%

C. Error Analysis and Confusion Matrix

Figure 2 displays the confusion matrix for the best-performing model (SVM on ResNet-18 embeddings). Most classes show strong diagonal dominance. Misclassifications occurred almost entirely between closely related sibling species in the same family—such as Brewer Blackbird vs. Rusty Blackbird, where both species have uniform dark iridescent feathers and similar bill geometry.

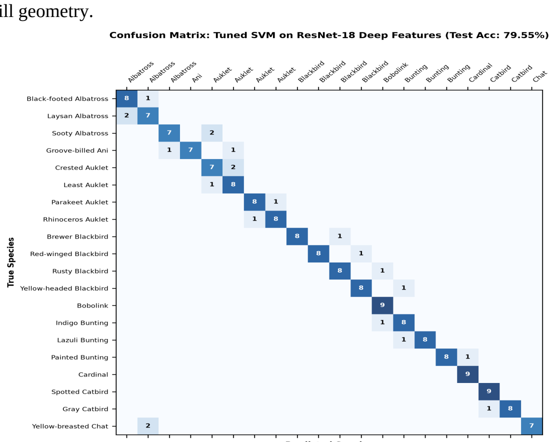

Fig. 2. *Confusion matrix for tuned RBF SVM on ResNet-18 deep embeddings (20 species, test set).*

VII. DISCUSSION

A. Why Classical Descriptors Plateau

Manual descriptors capture rigid geometric and color statistics without understanding semantic context. HOG tracks edge orientations, but a curved edge could belong to a wing, a beak, or a tree branch. LBP measures local contrast, but feather textures look similar across many passerines. When we concatenated these features into an 8,206-dimensional vector, the feature space became far too sparse for our 823 training samples, triggering the curse of dimensionality.

B. Why Deep Embeddings Succeed

ResNet-18 was pre-trained on ImageNet's 1.2 million diverse photos. Its convolutional layers have already learned hierarchical visual filters ranging from simple edges in early layers to complex morphological parts (eyes, feathers, beaks) in deeper layers. The 512-dimensional output forms a compact manifold where intra-class samples cluster tightly while different species are linearly separable.

C. Limitations

• We evaluated on 20 species; scaling to all 200 CUB classes will introduce more confusable species pairs and likely lower baseline accuracy.

• No data augmentation (crops, flips, color jitter) was applied during extraction.

• The ResNet-18 backbone was kept frozen; fine-tuning the convolutional weights on bird images would further improve accuracy.

VIII. CONCLUSION

We compared traditional hand-crafted feature engineering against transfer-learned deep representations for fine-grained bird classification on CUB-200-2011. While HOG, LBP, and color histograms plateaued at 36.36% test accuracy, frozen ResNet-18 embeddings paired with an RBF SVM reached 79.55% accuracy without any end-to-end backpropagation. This demonstrates that using pre-trained convolutional backbones as static feature extractors provides a practical, high-accuracy pipeline for fine-grained recognition in resource-constrained environments. Future work will explore backbone fine-tuning, data augmentation, and evaluation across all 200 CUB species.

IX. REFERENCES

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